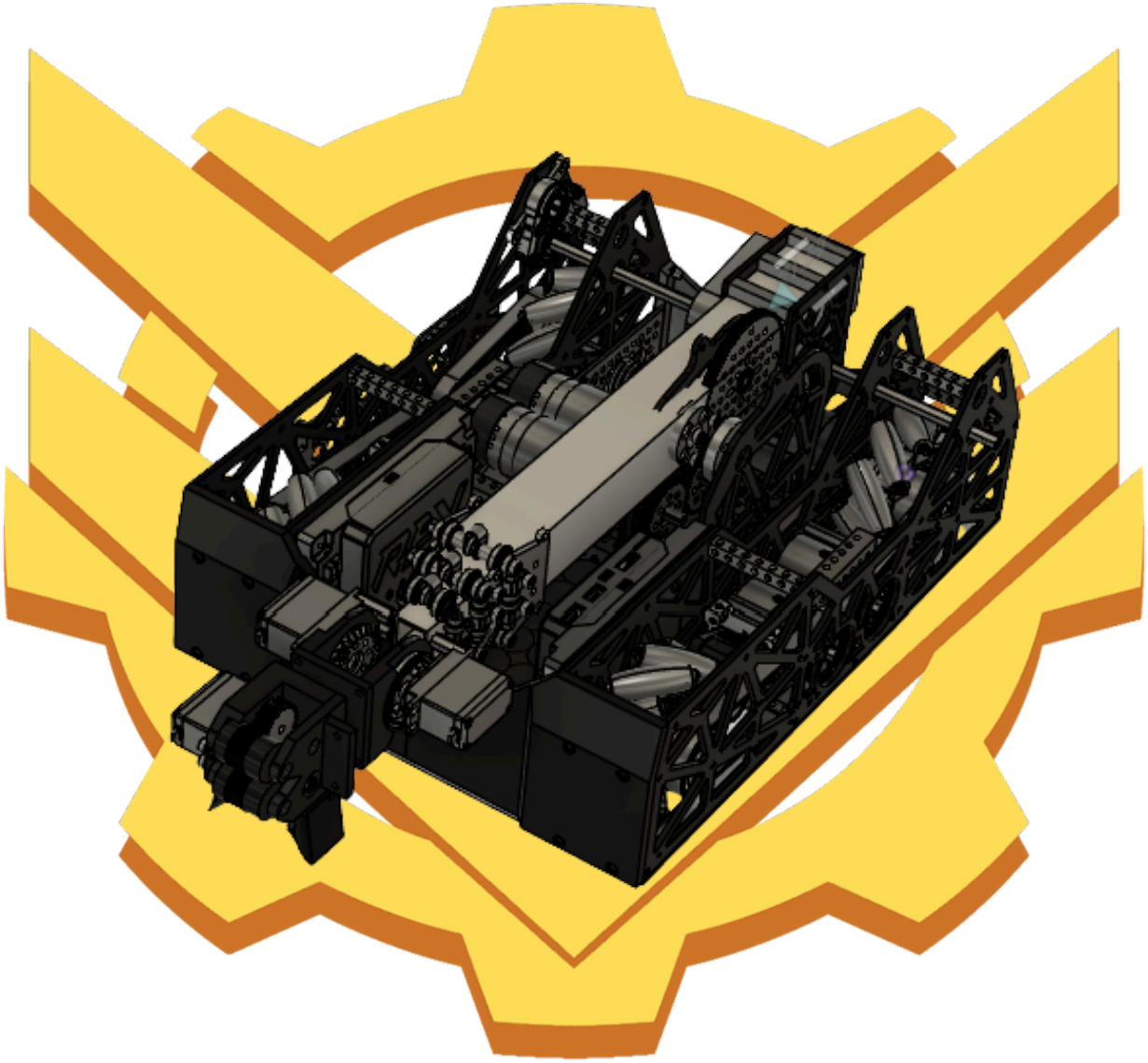
PARABELLUM

FTC TEAM 16031



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Team Introduction

Team PARABELLUM embraces a culture of innovation, collaboration, and excellence in robotics. Guided by the principles of **Gracious Professionalism** and **Coopertition**, we strive to create a supportive and dynamic environment where creativity and technical expertise thrive. Our mission is to push the boundaries of engineering, inspire others through meaningful outreach, and develop skills that empower us to make a lasting impact in the world of STEM.

Our Season Goals

This season, our goal is to create a system where every team member will learn an array of skills, while embodying **gracious professionalism.**

Develop Skilled and Confident Team Members

- Support continuous **skill development** for all members.
- Encourage stepping outside comfort zones, taking on new roles, and gaining hands-on experience.
- Create a **positive** and **welcoming environment** where questions and ideas are encouraged, allowing everyone to learn from each other.

Improve Team Efficiency and Organization

- Use task tracking software such as **ClickUp** to track deadlines, progress, and task responsibilities.
- Leverage **Discord** for team communication, meeting coordination, and updates.
- Maintain **GitHub** as a shared code repository for seamless collaborative coding.
- Keep a **Parts & Budget Spreadsheet** to track inventory, expenses, and financial planning.
- Hold Weekly Meetings (in-person and online) to monitor progress, solve problems, and share ideas together.

Expand Our Impact Through Outreach

- Ensure every member actively participates in outreach initiatives.
- Teach younger students coding and robotics using **LEGO Spike PRIME.**
- Connect with industry professionals to explore **STEM** careers and real-world applications of **FIRST**.
- Introduce **new communities** to FIRST and inspire a passion for STEM.

Gracious Professionalism

To us, Gracious Professionalism means competing and collaborating with **respect, kindness, and integrity** while aiming for the top. It fosters **selflessness**, **teamwork**, and **learning**, which creates an environment where everyone feels welcomed.

Mission Statement

Our mission is to inspire **innovation** and **excellence** through **teamwork**, **commitment**, and **creativity** which fosters mutual respect for our strengths.

Our Core Principles

Gracious Professionalism

Uphold respect, teamwork, and integrity, treating everyone—alliance or opponents—with kindness and respect.

Simple yet Efficient

Balance minimalism and effectiveness, reflected in both our robot design and team structure.

Adaptive Flexibility

Embrace new opportunities and roles, while continuously improving to meet new challenges.

Clear Communication

We meet twice weekly and communicate online to ensure consistent progress and team alignment.

Proactive Organization

Extensive use of CAD and component testing to help identify and address potential issues before they arise.

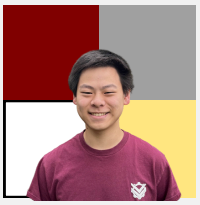
Meet The Team

Each member's **unique skills** and **goals** play a crucial role in our success. We each make **personal goals** for ourselves that help support our individual learning, while also assigning ourselves **robot components** to develop to support our team success. Through embodying **gracious professionalism** and **cooperation**, we strive to support each other in reaching our personal goals while also creating an efficient robot that will lead us to success in competition.

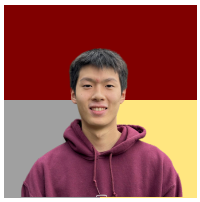
In addition, we utilize **flexible roles** rather than fixed roles compared to other teams. This means that no one person is limited to one role such as **programming** or **building**. Instead, everyone can do anything they choose to do, allowing us to grow in multiple areas and **try new things** rather than being bounded to one task. By sharing knowledge and encouraging others to try new things, we not only strengthen our robot, but our team as a whole.

We prioritize learning and growth by embracing flexible roles, allowing everyone to explore and try new things. The colors behind each member indicates their primary focus areas, while the subtitles beneath their names highlight the robot component they focused on the most for this season.

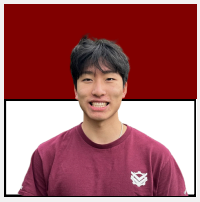
Building | **Drive Team** | **Coding** | **Outreach Planning**



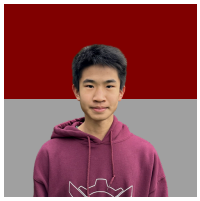
Simon Jin
Chassis, Box Tube
Personal Goal:
Develop leadership and machining skills by collaborating with others and exploring custom machining techniques.



Derek Wu
Hang, Intake
Personal Goal:
Strengthen problem-solving abilities by designing 3D-printed parts in Fusion 360 and working within resource limitations.

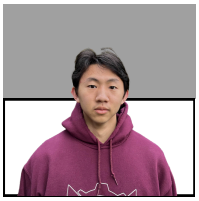


Leon Meng
Pivot
Personal Goal:
Cultivate a critical-thinking mindset by designing and engineering innovative solutions for the robot.

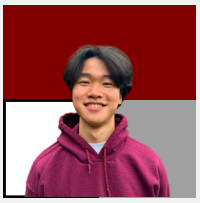


Lucas Wang
Roller Intake
Personal Goal:
Deepen my understand of the engineering design process through active review, testing, and continuous improvement

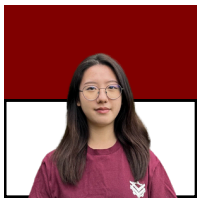
PARABELLUM '24-25



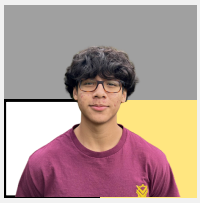
Weiwei Luo
Odometry, Autonomous
Personal Goal:
Expand my knowledge of pathing and coding concepts researching, seeking mentorship, and improving troubleshooting under pressure.



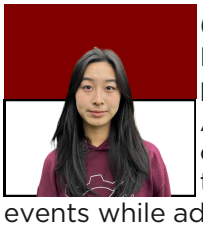
Matthew Lin
Autonomous
Personal Goal:
Enhance my ability to integrate different programs and collaborative in coding. while fostering an engineering mindset and improving my problem-solving skills through challenging myself.



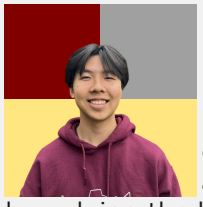
Bella Cheung
Outreach, Hang
Personal Goal:
Develop creativity and critical thinking by exploring innovative solutions and tackling engineering challenges.



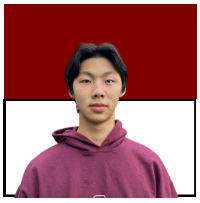
Julian Diaz
Object Detection, EP
Personal Goal:
Strengthen leadership skills by taking initiative in projects and fostering a supportive, motivated, and optimistic team environment.



Claire Huang
Passive Intake
Personal Goal:
Apply leadership and communication skills to organize outreach events while advancing my knowledge of designing and 3D-printing robot components.



Ethan Zhang
Roller Intake
Personal Goal:
Focus on personal growth, technology, and professionalism by valuing the learning process over competition results.



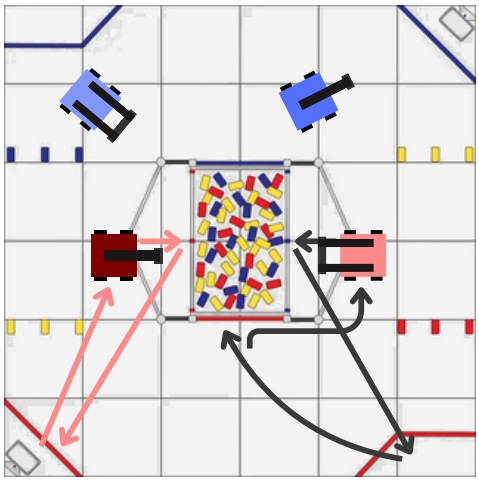
Vincent Xiao
Chassis, Passive Intake
Personal Goal:
Advance my CAD and robot programming skills by exploring online techniques and seeking mentorship within FIRST.

Game Strategy

In this year's game, the two primary scoring methods are **chambers** and **baskets**. Since **speed** and efficiency were our main deciding factors, we analyzed **average cycle times** and **points per second** to determine the optimal strategy.

	Average Cycle Time	Points Per Second
High chamber (10 Points)	14.4s	0.69
High bucket (8 Points)	10.4s	0.77
Preloaded High Chamber (Autonomous)	6.2s	3.22
Preloaded High Bucket (Autonomous)	5.8s	2.76

Using the table above, we can see that the **high bucket** has a **faster average iteration time** and scores more points per second. Thus, we prioritize **high buckets** over high chambers and other scoring methods.

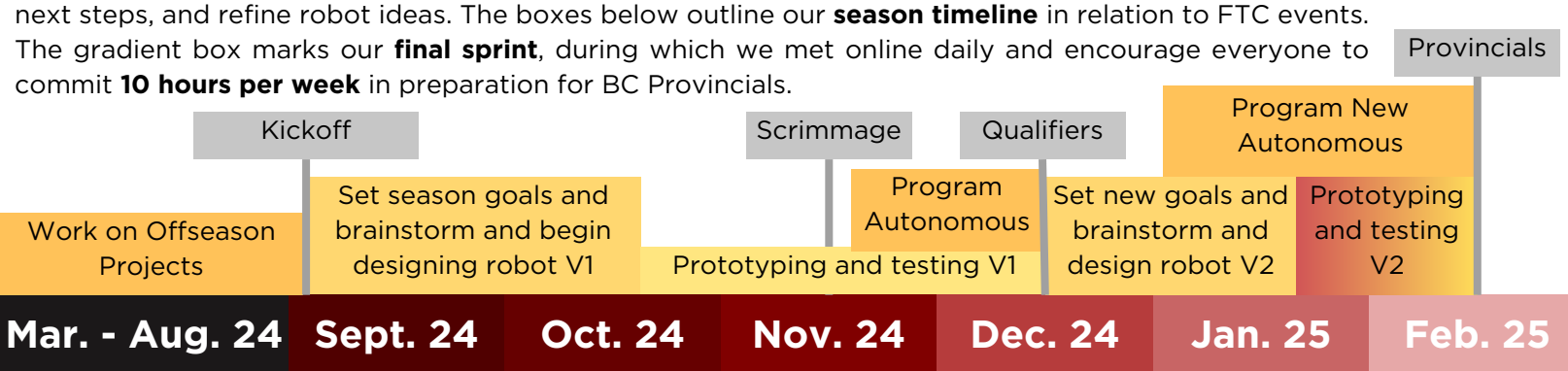


The diagram above maps out the paths we take when scoring in red alliance high buckets and high chambers with pink and black arrows, respectively.

Autonomous			Tele-Op	Endgame
Put a sample into the high bucket.	Collect and place three samples onto high basket.	End autonomous period by parking in submersible zone.	With an 8s cycle time, we estimate that we can complete ~13 cycles.	Complete second stage hang on chamber.
7 Seconds	**18** Seconds	**5** Seconds	**110** Seconds	**10** Seconds
8 Points	**24** Points	**3** Points	**104** Points	**15** Points

Season Timeline

Every week, our team meets in person on Sundays and online on Wednesdays to track progress, plan next steps, and refine robot ideas. The boxes below outline our **season timeline** in relation to FTC events. The gradient box marks our **final sprint**, during which we met online daily and encourage everyone to commit **10 hours per week** in preparation for BC Provincials.



Engineering Design Process

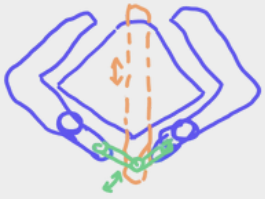
For every part of our robot, we follow a structured four-step process to ensure top-quality components: planning, **designing, developing, and testing.**

Planning

Step 1

Planning our ideas allows us to sift through ideas and determine our goals and how to accomplish them. We do this by meeting together to discuss multiple different ideas, noting down their individual **pros** and **cons**. We generally do our planning using whiteboards and online meetings with all our team present, so that we can hear from all perspectives and that everyone is on the same page.

Claw Intake Example



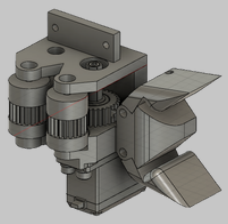
Finite State Machine Example

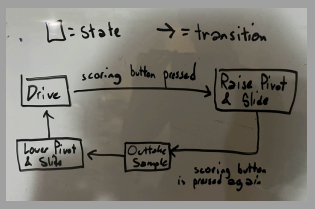
When planning our Finite State machine, we wrote down our goals for the program. e.x. simple, easy for the driver, easy to refine

Designing

Step 2

We develop a flexible prototype that isn't yet implemented on the robot. This allows us to identify potential issues without committing **excessive time** and **resources**. For hardware, this usually means3D modeling, while for software, we create a diagram outlining our program's intended **functionality**.

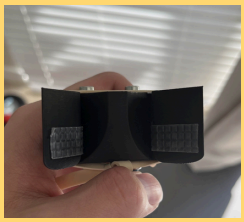




Developing

Step 3

After finalizing our drafts, we move on to building a functional prototype that can be used on our robot. Having a real-life version helps us identify **flaws** or **areas for improvement** in real-world scenarios, ensuring our final version meets our standards.

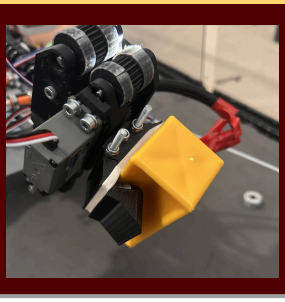


Continued to develop finite state machines as we added more mechanisms that needed programming.

Testing

Step 4

The final step is testing. This is when we put our prototypes to the test, and see if they can meet our performance expectations. This can include repairability, durability, and the ability to fulfill its purpose. If something is wrong, we repeat the design process until we find a suitable solution.



Tested, tuned, and improved the code by using the FTC Dashboard and Android Studio

Partnering With Other Teams

When designing our robot components, we collaborated with six other FTC teams, sharing ideas and enhancing each other's designs.

#23432 PMR INFERNO	#27560 PMR IGNITE	#24230 PMR PHOENIX	#14864 WWW	#11228 OverClucked Bots	18763 Texpand
Port Moody, BC, Canada	Port Moody, BC, Canada	Port Moody, BC, Canada	Wilmington, MA, USA	Zeeland, MI, USA	Cape Town, WC, South Africa

CENTERSTAGE CHAMPIONSHIP WINNING ALLIANCE

Business, Risk, & Sustainability Plan

Goal & Strategies

Our goal for this season is to keep our **expenses** within our budget. We were able to fundraise through events and sponsorships like Protocase.

Fundraising at Community Events

- Hosting **workshops** and booths at events like the Canada Day Parade to raise awareness and attract donors.

Applying for Grants and Sponsorships

- Applying for **grants** from organizations such as **BC Hydro**
- Collaborating with **local businesses** for funding and support.

Sustainability Plan

Our team strives to create a **lasting impact** in our community through inspiring others with the opportunity to experience FIRST and its multifaceted benefits.

Recruitment & Retention

Hosting **open houses** and **workshops** to recruit new members and develop their skills in coding, outreach, and building.

Knowledge Transfer

Documenting processes, code and designs for future use in platforms like **Google Drive** and **GitHub**.

Community Partnerships

Collaborating with **local schools** and **businesses** to foster interest in STEM and raise funding.

Financial Stability

Seeking **multi-year sponsorships** through applying for grants annually, and launching fundraising campaigns to cover expenses.

Expenses & Income

Category	Expenses
Registration	$1600
Robot Supplies	$3300
Tools	$1150
Food	$270
Outreach	$30
Total	$6350

Category	Income
Fundraising	$1000
Sponsorships	$4350
Total	$6350

In-Kind	Expenses
In-kind sponsorships	$500
Part donations	-$600
Total	$-100

Risk Management Plan

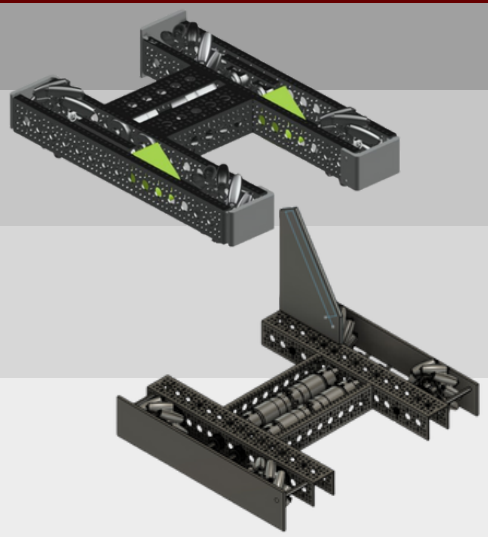
When designing a new robot component, we identify and analyze **potential risks** to develop effective solutions. We brainstorm and evaluate potential weaknesses and challenges. Through accessing the **likelihood** and **impact** of each risk, we are able to determine mitigating strategies for both hardware and software. An example table of risks is included below:

Example risks	Likelihood	Impact (Result of Risk)	Mitigation Strategy
Use of Custom Designed Parts	Likely	Very High (Difficult to replace, takes far longer to perfect, more expensive)	• Start designing early in the season • Create many prototypes that go through many tests and redesigns • CNC parts instead of 3D printing
Reliance on Automated Pickup Sequence	Highly Likely	Low (manual override for pickup)	• Driver practice for manual override • Several options (Roboflow, external repositories, openCV)

Off-Season Projects

The goal of off-season projects is for members to **freely explore** various areas of robotics knowledge by working on a project of interest. These passion projects will teach us valuable lessons that we can apply during the season, including new **drivetrains**, **attachments**, and **code**.

New Chassis

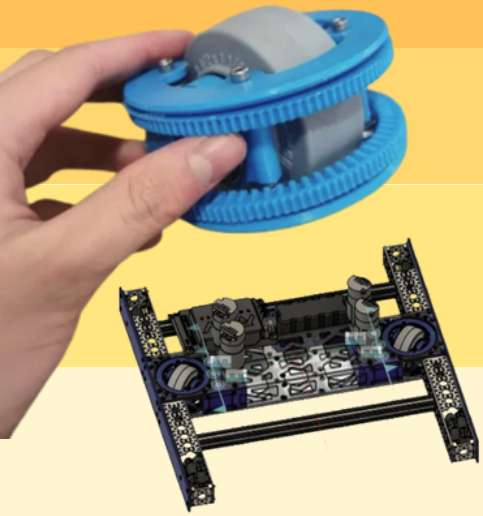


What is it? This season, we chose a new chassis design that is **simple** and **effective**. It features parallel plates connected by standoffs and bumpers, with motors pushed through the plates to save space.

Why did we decide to switch our chassis? These changes made the robot quicker and more compact. The packaging improved, shrinking the width of the chassis from 14 to 11 inches. Furthermore, a **lower centre of gravity** enhanced stability.

What did this project teach us? We learnt new **machining techniques**, including the use of a drill press. Throughout the process, our CAD skills improved, reinforcing the idea that a simple solution can be better than a complex one.

Swerve Drive



What is it? Our swerve drive uses a **differential** to transfer power for both **rotation** and **movement**, allowing two wheels to spin 360 degrees for full omnidirectional control.

Why did we decide to switch our driving method? Four omni wheels keeps the robot **balanced**, making it more **power-efficient** than a mecanum drive while maintaining the same speed in all directions. This design also **increased** our **top speed** for faster movement.

What did this project teach us? Through this process, we learned how to program and build a swerve drive. We researched differential systems for potential use in a 2-axis wrist, which improved our 3D-printing techniques. This experience reinforces the idea that complex designs are not always best option.

Ultrasonic Sensor

What is it? We experimented with ultrasonic sensors, which are distance sensors that measure distance using sound waves.

Why did we experiment with it? Why did we not use it? One of the reasons we wanted to experiment with sensors was to have a backup for our odometry during autonomous. During our testing, we found many issues with the design, which is why we decided not to use it on our robot. One of the issues was a minimum distance requirement. If the distance was less than 1cm, then the sensor would read it as the maximum of 3 meters.

What did this project teach us? We learned how to code the sensor from scratch, as well as the intricate details about how sensors worked. Furthermore, the experience taught us that we must carefully decide what to utilize because there are a variety options.

Outreach

This season, our team has inspired others and connected with thousands. We promoted **STEM** through **outreach events**, including the Canada Day Parade to workshops at schools and libraries. We support the FIRST community by **collaborating** with different teams and hosting STEM events. Engaging with **industry professionals** has also deepened our understanding of the **real-world application of robotics**.

Outreach & Service Plan

At the start of the season, we planned the types of outreach events we wanted to attend and host. Below, we've listed each category along with the events we actually participated in.

Introducing Students to FIRST & STEM

Hosting workshops at schools and libraries for younger students to introduce them to simple robotics and FIRST.

- Kwayhquitlum Middle School
- Coq. Library workshop
- Yukon Crescent Children's Centre
- Kitchener Elementary School

Helping Other Robotics Teams

Aiding other robotics team through supplying materials, giving advice, and sharing our knowledge.

- Number Plates for other FTC Teams
- Kick-off Presentation
- Aiding School Clubs

Giving Back to Our Community

Engaging with our community by offering support and spreading awareness about FIRST.

- Berry Picking
- Canada Day Parade
- Street Clean-Up

Connecting with Professionals

Learning from industry professionals about their field of STEM, while also making and sharing connections we found with their workforce and FIRST Competitions.

- Porsche Meeting
- Seaspan Shipyards Visit
- BS99
- EA Industry Professional
- Chernoff-Thompson Architects
- SCP Consultants

Outreach Total Stats

FTC Teams Helped	Outreach Events
18	16
Outreach Hours	**Materials Loaned**
~260	~$600

People Impacted & Reached

16 000+

Planning Events

Each event is carefully structured to ensure **maximum engagement** and learning. We first identify the **target audience,** whether young students, FIRST teams, or professionals, and tailor our activities accordingly. For younger students, we create hands-on activities and storytelling-based lessons, while FIRST events focus on strategy discussions and technical demonstrations. **Industry outreach** involves preparing in-depth presentations that highlight robotics applications in engineering, automation, and design. Every event includes a **post-outreach evaluation**, allowing us to refine and improve our engagement strategies.

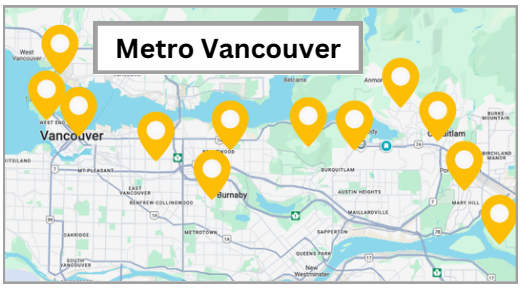
Making Connections

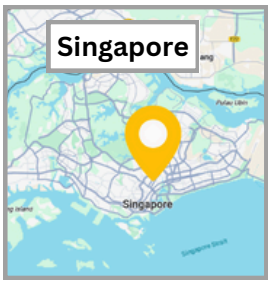
We maintain connections with schools and daycares by providing teachers with follow-up **robotics resources**, ensuring students can continue learning. For newer FIRST teams, we have continually offered mentorship, **CAD tutorials** and **strategy discussions**. Our industry connections remain active through **email follow-ups** and **progress updates**, allowing us to seek advice and strengthen professional relationships. This ongoing engagement ensures our outreach has **lasting impact** beyond the initial interaction.

Outreach Cont.

Outreach Map

This year, we engaged with communities across **Metro Vancouver**, helping introduce people to **FIRST** and **STEM**. We also connected with professionals from **Vancouver**, **Italy** and **Singapore**.





Community Outreach

Canada Day Parade

Participated in the local **Canada Day Parade** to represent our team and FIRST, engaging with over **15 000 attendees** to promote STEM and robotics. This event allowed us to showcase our robot to many of whom learned about robotics, STEM, and FIRST for the first time.

Aiding School Clubs

We supported the **PMSS F1 Club** and **PMSS robotics** teams with 3D printing and STEM-related design, inspiring others through hands-on learning and encouraging **creativity** in **problem-solving** and engineering.

Number Plates for Other FTC Teams

We created custom number plates for 20 teams in our region, fostering a culture of collaboration and recognition within the FIRST community. It also gave newer teams the opportunity to receive aid.

Kick-Off Presentation

We delivered a presentation at this season's kick-off event, engaging an audience of **100 students** and **mentors** to discuss **team strategies** and share **insights** about the new game. This opportunity introduced our team to other FTC teams in our region, helping our community know more about us.

Berry Picking Fundraiser

We organized a berry-picking outing as a fun **team bonding** activity while also supporting a local farm. To help reduce the stress on our team's budget, we sold blueberries and other equipment like a canopy to **raise money** and share our work and information about FIRST with consumers.

Street Clean-Up

To expand our team's presence and promote FIRST within our community, we participated in the '**Adopt-a-Street**' program. Through this initiative, we committed to maintaining designated streets in our neighbourhood, marked by street posts with our team name, by conducting **regular clean-ups**, and showcasing our team while also fostering a **clean environment**.

Industry Professionals

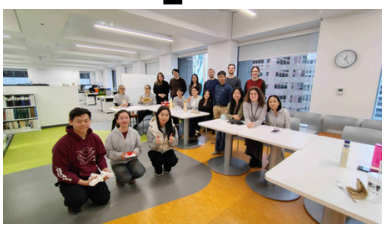
Porsche Meeting

We held an outreach session with Porsche's testing department, sharing insights on robotics and gaining **valuable feedback** on **testing methodologies**. The interaction inspired our team to improve our testing system and practices, and consider industry applications of our work.

BS99

We attended an **invite-only** event where members of our team interacted with and learnt with engineers regarding **computational insights** and **industrial expertise**. Our team, alongside Furious Frogs, presented about FIRST, FTC, gracious professionalism and the engineering design process. **Dr. Erhan**, Professor at SFU's School of Interactive Arts was the keynote speaker.

Chernoff-Thompson Architects



We explored the real-world applications of 3D printing in construction and urban design. By understanding how professionals use **iterative design** and **prototyping**, we refined our own approach to testing and improving our robot's mechanical components. We showcased our 3D printed parts to the principal architects, associate architects and **certified professional.**

Keeping Contact

We keep in touch with mentors and industry experts through follow-up **emails** and **project updates**. Some follow-up discussions with **Porsche Engineering** has allowed us to have further discussions about opportunities like designing a competition for their **summer robotics camp**. These relationships not only help us improve our own engineering skills but also open doors for **future collaborations** and **mentorship opportunities**.

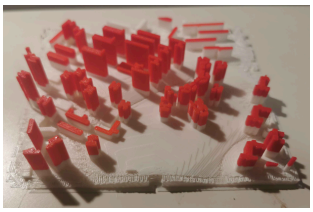
Seaspan Shipyards Visit

We had the privilege of touring **Seapan's shipyard** with its own **Director of Operations**. We learned about the assembly process for many Canadian ships, including the **JSS1**, the longest naval vessel ever built in Canada. This visit provided valuable insights into the **shipbuilding industry**.

EA Front-End Developer

We met with an **EA Front-End Developer** and presented our robot and introduced them to FIRST. We gained valuable insight regarding our current activities and how it relates to **professional fields** which many of our team members may want to pursue in the future.

SCP Consultants



We had the opportunity to have a video conference with **CP Consultants Pte Ltd (SCP)** from **Singapore**. We examined how automation and robotics are transforming city **infrastructure** and **development**. They shared insights on how robotics is used in large-scale projects for precision and efficiency, reinforcing the importance of CAD-based modeling and simulation in engineering applications.

Applying Information

Our industry outreach events provided valuable insights that influenced our **engineering approach**. Vancouver architecture professionals emphasized the importance of material selection in 3D printing, leading us to experiment with stronger filaments for our robot parts. Porsche engineers stressed the need for **structured testing**, inspiring us to implement a detailed **testing log** for tracking performance. By applying these industry insights, we improved both our **robot's functionality** and our **engineering process**.

Educational Outreach

Kwayhquitlum Middle School

Hosted an **interactive workshop** for middle school students, introducing them to **robotics** and **programming** basics. This initiative aimed to inspire younger students to explore STEM and consider joining FTC in the future.

Coquitlam Library Workshop

Organized a public robotics workshop at the Coquitlam Library to teach children and families about **building** and **coding robots**. This event reached a wide audience and helped demystify robotics for the general public.

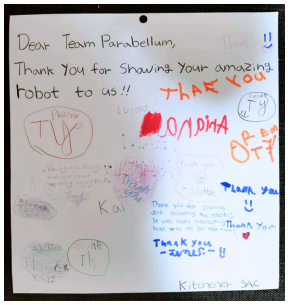
Kitchener Elementary School

We conducted a hands-on **programming** and **building** session to encourage early STEM exploration in **elementary students** with lego robotics and simple block coding.

Yukon Crescent Children's Centre

We encouraged early STEM exploration to **preschool** children (under age 6) to lego robotics and simple **block coding** through basic **Lego engineering** and **motion mechanics**.

Collecting Feedback



Through our middle school and library robotics workshops, we collected **feedback** from teachers and students on **learning challenges** and **engagement strategies**. Educators highlighted that real-world applications helped keep students engaged, leading us to incorporate **practical problem-solving** challenges into our curriculum. Adjusting our approach based on feedback has helped us make robotics more **accessible** and interactive for younger learners.

Mentors

Our mentors are essential to our growth, embodying **Gracious Professionalism** by guiding us through challenges, encouraging **collaboration**, and refining our skills in engineering and teamwork. Their support fosters a **positive learning environment**, helping us adapt, improve, and push the boundaries of **innovation**.

Kevin Zhou
Provided guidance on **robotics concepts**, helping the team refine designs and technical decisions.

Trixie Wiriahardja
Supported the team in **improving organization** by offering advice on task management, schedules, and responsibilities.

Stephanie & Eric
Offered valuable feedback on the **engineering portfolio**, helping improve clarity, structure, and technical documentation.

General Design

This season, we decided to follow 2 design tenets that we determined to be

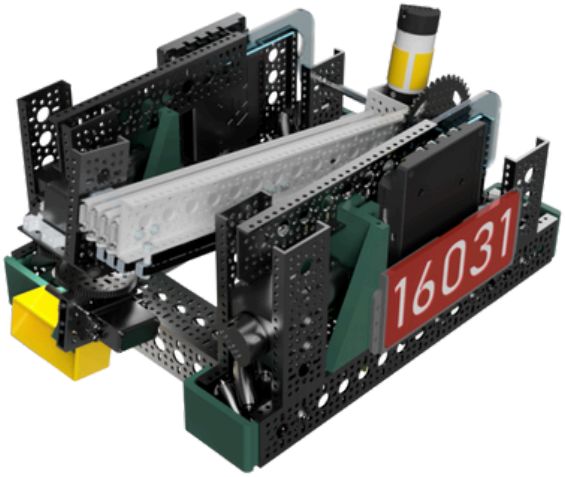
Elegance

Designing a robot with 3D-printed, machined, and pre-made parts for minimalism, durability, and efficiency

Effiency

Integrating mechanisms to optimize cycle times, maximizing efficiency and points.

Version 1



This was our robot from the start of the season through the qualifiers. After the qualifiers, our team decided to redesign it to address the flaws we discovered.

Description

- Gobilda 12.0″ x 16.5″ parallel plate drive train

Strengths

- Rigid chassis are able to endure collisions
- Big and simple designs are easy to repair

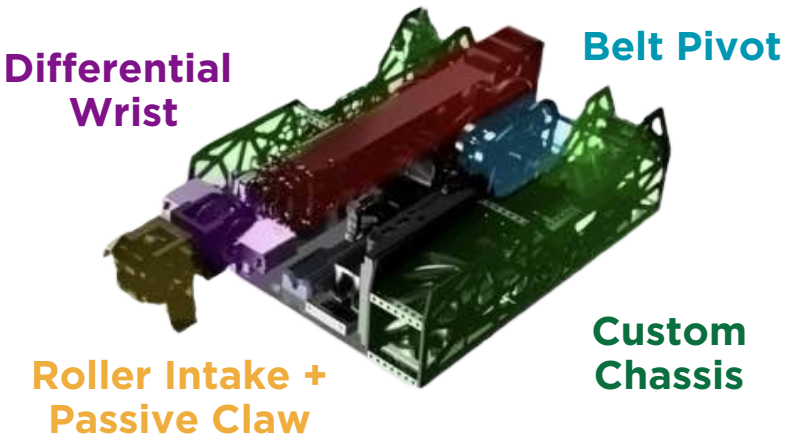
Weaknesses

- Viper kit extension lacked rigidity
- Sprocket and chain had much backlash
- Claw intake was slow to pick up samples and limited to ground
- 1.5 times as heavy as V2
- Hang as a separate mechanism contributed weight and took power away from other mechanisms

What We Learnt

- After the provincial qualifier, we learned that there is a need for more rigidity and speed
- Even things like our drivetrain could be stronger and faster, so we decided to improve every aspect of our robot

Version 2



Description

- Custom 11.5″ in x 15.5″ machined drivetrain

Strengths

- Box tube improves rigidity and halves weight.
- Herringbone gears reduce backlash.
- Roller intake and color sensor speed up intaking.
- Passive claw and differential wrist grab from ground and wall.
- Integrated pivot hang reduces weight.
- Pocketed chassis and higher RPM cut weight and boost speed.

Weaknesses

- Chassis vulnerable to heavy collisions
- Compact intake mechanism vulnerable to collisions
- Intricate mechanisms make repairing difficult
- Compact robot makes wiring more difficult

What We Learnt

- Custom chassis outperforms Gobilda plates by tailoring to design needs (e.g., pivot holes for extending arm).
- Allow ample time for testing and refining complex mechanisms.
- Recognized the strengths and limits of custom-machined parts.

Scoring Elements

Box Tube

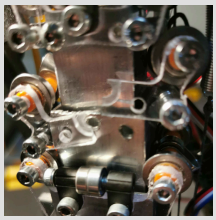
After the problems our slide encountered at qualifiers, we decided to switch to a **custom box tube design** that is seen in FRC that offers better **rigidity** and **speed**. It is a series of **telescopic aluminum tubes** supported with an outside bearing bracket and an inside bearing bracket. Gives more **rigidity** compared to a typical linear slide from support on 8 points and 4 sides, and is **faster** since it uses 2 motors and cascading spools

Problems

When we first tried creating a box tube, there were many problems as we had not done something similar before. The 3D-printed parts **lacked strength**, the metal mount was too heavy, the design wasn't compact, and the extension was too short.

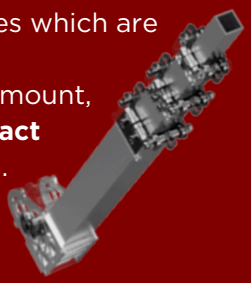
What We Tried?

- Tuned 3D-printed filaments for strength.
- CNC polycarbonate plates broke.
- To CNC metal was too expensive, difficult, and would be too heavy.



Solutions

Our final design features CNC carbon fiber bearing plates which are much **stronger**, a thinner pocketed mount, and a more **compact** 4-stage extension.

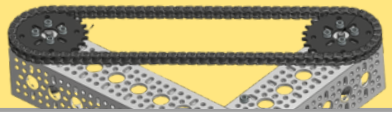


Pivot

For our pivot, we have a **1:6 ratio** gearbox connecting to two 435 RPM motors, giving us a stall torque of 224.4kg/cm of torque and an average pivot speed of 72.5 RPM, a significant improvement in torque and speed from our qualifier robot. Our **herringbone** gears have **minimal backlash**, offering **superior precision** and **stability.**

Problems

Our pivot at **qualifiers** was too **slow**, and **unstable**. After investigating the cause, we determined that it was due to **backlash** in our sprocket and chain, and a lack of pivot power.

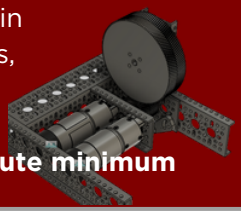


What We Tried?

- Tried an **experimental capstan drive**, which is a 0 backlash speed reducer powered by rope
- Larger tensioners, and two pulleys to reduce backlash and tensioning issues

Solutions

Decided on **herringbone** gears for our final version. With at least 2 angled gear teeth in contact at all times, **skipping** and **backlash** is kept at an **absolute minimum**

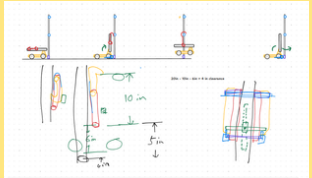


Hang

To save our motors for other mechanisms, we created a 3D printed **passive hanging** that uses the pivot.

Problems

Our first hang iteration restricted space on the tube. We made it long enough so it could reach the bar, but the hooks were too long that the robot couldn't lift itself.

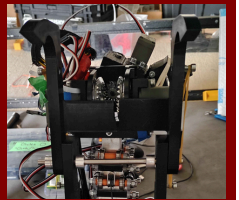


What We Tried?

- **2 stage hang** attached by a magnet
- Various heights from calculations depending on slide positions
- Different release mechanisms to ensure hook could release at the right time

Solutions

Our final iteration is a **3 stage hang**, including the tube slide. We made it **foldable** so it did not restrict space on the tube slide. It is released using **elastic bands**, a **blocker**, and a **servo**.

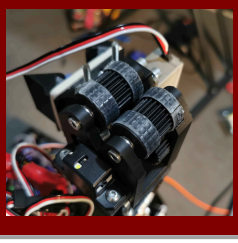


Intake Elements

During qualifiers, our claw intake had many problems we found afterwards, so we split it into 2 different scoring components that allow us to be multifaceted within the game.

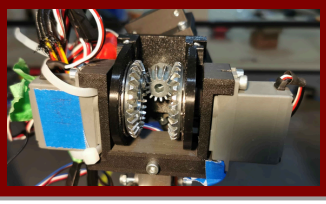
Roller Intake

Our roller intake picks up samples from the field or truss to score in high buckets, with **single-sample clearance** to prevent penalties. Grip tape ensures reliable pickup, and the lightweight design is mainly 3D-printed. A color sensor detects samples and **triggers an RGB light**, helping the driver see intake activity.

Problems	What We Tried?	Solutions
Our claw mechanism at qualifiers lacked **accuracy** and **precision**, was **slow**, and required specific sample orientation, proving unreliable and needing replacement.	• Tried different roller intake designs • Tried different tapes and **materials** on the rollers • Tested **color sensor** and **touch sensor** - Color sensor was more effective overall	Our roller intake reduces penalties, increases speed, and lowers **precision** needs, benefiting the driver. The light indicator improves **visibility** during intake. 

Differential Wrist

Connected to the roller intake, we have a differential wrist, which gives our mechanism versatility to pick up samples oriented in **any direction.**

Problems	What We Tried?	Solutions
Our claw mechanism had little to no **range of motion**, heavily limiting which samples could be picked up depending on their **orientation**.	• Creating a mount to create **stronger support** between two servos • Trying **3D-printed** gears • Testing different gear ratios • Finalizing on a **2:1** gear ratio since there is more **range, compactness and speed**	We created a differential wrist connecting our box tube and intake roller. This allows the roller to orient for picking up samples in **any orientation.** 

Passive Intake

Our second scoring component is our passive intake, which will predominantly be used to score specimens. This design works ultimately as an **intake and outtake**, a claw that allows our robot to run into a specimen, and attach it. Then when scoring, our robot just needs to run into the truss, and the specimen will detach.

Problems	What We Tried?	Solutions
Using the claw to get specimens required additional precision from the driver, which resulted in **longer cycle times** for specimens.	• Using different materials to increase grip of the claw - Rubber bands, cat tongue tape • Different **angles of approach** when releasing a specimen, and different angles for the claw • Trying different claw thicknesses	With our passive intake, we drive into specimens to grab them and into the truss to score them, reducing **cycle time** and **driver imprecision.**

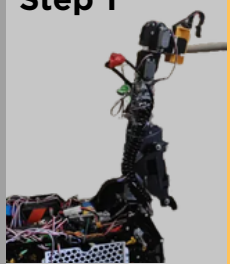
Automation

The game this year comprises many **complex precise movements**, in multiple dimensions. This can make it very difficult for the driver to perform tasks at **efficient speeds** manually.

Rather than having our driver press multiple buttons to complete an action, we made multiple **one-button sequences** that will complete entire actions, making the driver's job easier. Below is a visualization of one of our one-button sequences, a one-button sequence to score on the high chamber.

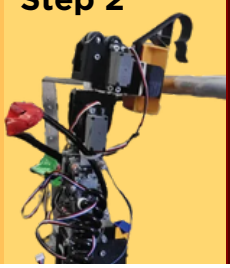
Automatically rotate the pivot and raise the box tube to the precise scoring position dictated by **encoders**

Step 1



Using the **coordinates** on the field, drive until specimen is positioned correctly to score

Step 2



Rotate the arm down to release the specimen from the passive claw, retract box tube, and lower pivot to ensure low center of gravity.

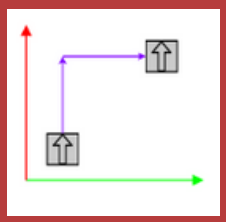
Step 3

The **automation** in our robot is achieved through the use of many **sensors** and **software programs**. While the functions that are automated seem simple, they required complex **computer vision software** and **vector calculations** to achieve. Object detection and pickup is possible because of the **Limelight** and precise drive coordinates and automated driving at high speeds is possible because of **Odometry**.

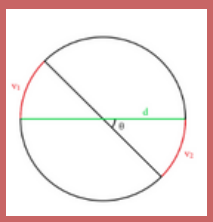
Odometry & Localization

FTC odometry uses sensor-equipped wheels to track a robot's position with ~1 cm accuracy. Movement is analyzed in **time steps**, calculating **angle changes** from encoder differences and correcting for horizontal shifts due to rotation. Position changes are then converted to field coordinates using **trigonometry** to ensure **precise** autonomous navigation.

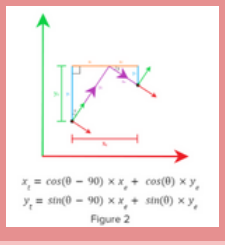
Step 1: Calculate horizontal/ vertical change



Step 2: Calculate angle change



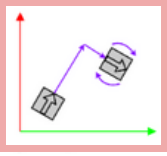
Step 3: Calculate h-error



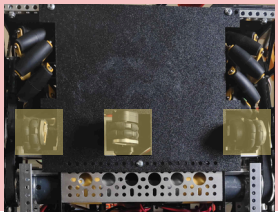
$$x_t = cos(\theta - 90) \times x_e + cos(\theta) \times y_e$$
$$y_t = sin(\theta - 90) \times x_e + sin(\theta) \times y_e$$

Figure 2

Step 4: Translate to coordinate plane



$$x_t = cos(\theta - 90) \times x_e + cos(\theta) \times y_e$$
$$y_t = sin(\theta - 90) \times x_e + sin(\theta) \times y_e$$



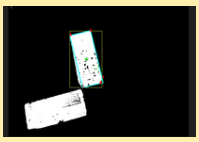
Our robot uses 3 Gobilda Odometry pods: 2 mounted vertically on either side of the robot (Vr and Vl), and one horizontally (H).

Object Detection

Our first scrimmage showed that **manually adjusting** the robot to pick up samples took more than half our cycle time. To improve efficiency, we prioritized **object detection** to assist the driver and speed up the process. When a button is pressed, the robot detects samples in the submersible based on **color**, calculates their **angle** and **position**, and adjusts itself for pickup. To achieve this, we compared two solutions: a Limelight 3A and a webcam. We ultimately chose the **Limelight 3A** due to its superior qualities.

Step 1: The Limelight captures an image on button press.

Step 2: Apply Vision Pipeline to isolate colour.



Step 3: Determine target contour.

Step 4: Calculate position, and angle of target contour.

Step 5: Adjust robot, arm, and intake using Limelight data.

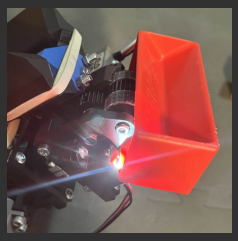
Automation Cont.

RGB Colour Indicator

When picking up samples from the submersible, the intake would sometimes obscure the sample from the driver's view, making it difficult to see the colour of the sample being picked up. Therefore, we attached a colour sensor on our roller intake, and a RGB Colour Indicator on our chassis so that we could see the colour of the sample being picked up. This way, we can avoid penalties for accidentally collecting samples from the opposing alliance.

Colour Sensor

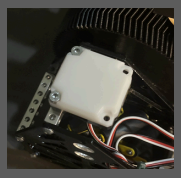
We designed a 3D mount for a color sensor on our roller intake to detect the color of the samples being picked up. The detected color is then displayed on the color indicator.



Colour Sensor on Roller Intake with Red Sample

Colour Indicator

We created a 3D mount for a colour sensor on our roller intake, so that it could detect the colour of the sample being intaked, which will then be outputted on the colour indicator.



Colour Indicator With No Sample

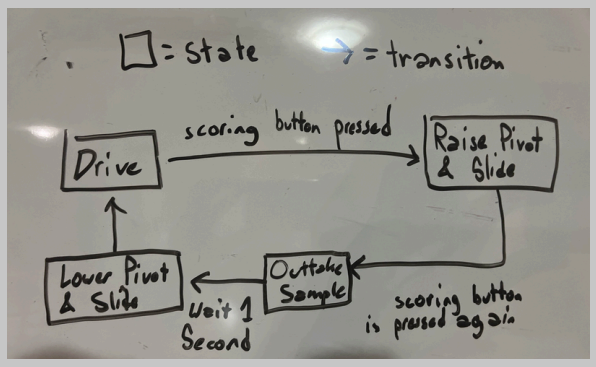
With Blue Sample

With Yellow Sample

With Red Sample

Finite State Machine

Our finite state machine connects all of our mechanisms into one program, all being controlled through button presses and other event triggers. Using states that control multiple robot components at a time, and transitions that rely on timers This system organizes our programs and ensures coordination by linking actions to corresponding triggers, allowing for smooth transitions and enhanced performance.



Here is an diagram of how the finite state machine works with the scoring of samples in bucket:

1. The robot starts in the drive state, where the pivot and slide are in a lowered position while the robot drives.
2. When the designated scoring button is pressed, the pivot and slide are raised to a position where the sample can be released from our outtake.
3. When the designated scoring button is pressed again, the outtake drops the sample into the bucket.
4. After a one-second wait to ensure the sample is clear, the pivot and slide return to their lowered position, and the robot resumes its original drive state, ready to activate other states.

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